Light

GCSE Computer Science — Boolean Logic (Set 1)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A logic gate takes two inputs, A and B.
The output is 1 only when both A AND B are 1.
What type of logic gate is this?

- ○ **A.** An AND gate
- ○ **B.** An OR gate
- ○ **C.** A NOT gate
- ○ **D.** A NOR gate

Q2 (1 mark)

A security system has two sensors, P and Q.
The alarm sounds if sensor P OR sensor Q detects movement.
When will the alarm sound?

- ○ **A.** Only when both sensors detect movement
- ○ **B.** Only when sensor P detects movement
- ○ **C.** When at least one sensor detects movement
- ○ **D.** Only when neither sensor detects movement

Q3 (1 mark)

A NOT gate has a single input X.
What does a NOT gate do?

- ○ **A.** It outputs the same value as the input
- ○ **B.** It outputs 1 only when the input is 1
- ○ **C.** It outputs the opposite of the input
- ○ **D.** It outputs 0 regardless of the input

Q4 (1 mark)

An AND gate has inputs A = 1 and B = 0.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** Cannot be determined

Q5 (1 mark)

An OR gate has inputs A = 0 and B = 1.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 0.5
- ○ **D.** Undefined

Q6 (1 mark)

A NOT gate has input X = 0.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** -1
- ○ **D.** No output

Q7 (1 mark)

A truth table is created for a logic circuit with 2 inputs.
How many rows are needed to show all possible input combinations?

- ○ **A.** 2 rows
- ○ **B.** 3 rows
- ○ **C.** 4 rows

- ○ **D.** 8 rows

Q8 (1 mark)

A car will only start if the key is inserted AND the driver presses the start button.
The key is inserted but the button is not pressed.
Will the car start?

- ○ **A.** Yes, because the key is inserted
- ○ **B.** Yes, because one condition is met
- ○ **C.** No, because both conditions must be met
- ○ **D.** No, because the button overrides the key

Q9 (1 mark)

A door unlocks if a valid keycard is used OR a correct PIN is entered.
A person enters the correct PIN but does not have a keycard.
Will the door unlock?

- ○ **A.** No, because both are required
- ○ **B.** No, because a keycard is always needed
- ○ **C.** Yes, because at least one condition is met
- ○ **D.** Yes, because PIN always overrides keycard

Q10 (1 mark)

In Boolean algebra, the expression for 'A AND B' can be written as:

- **A.** A + B
- **B.** A ∧ B
- **C.** A − B
- **D.** A / B

Q11 (1 mark)

In Boolean algebra, the expression 'A OR B' can be written as:

- **A.** A × B
- **B.** A ∧ B
- **C.** A ∨ B
- **D.** A ÷ B

Q12 (1 mark)

In Boolean algebra, 'NOT A' can be written as:

- **A.** A + 1
- **B.** A − 1
- **C.** Ā (A with a bar over it)
- **D.** A × 0

Q13 (2 marks)

Evaluate the Boolean expression: A AND (NOT B)
Where A = 1 and B = 1.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** NOT 1

Q14 (2 marks)

Evaluate the Boolean expression: (A OR B) AND C
Where A = 0, B = 1, C = 1.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 3
- ○ **D.** Undefined

Q15 (2 marks)

A truth table is created for a logic circuit with 3 inputs: A, B, and C.
How many rows are needed to show all possible input combinations?

- ○ **A.** 3 rows
- ○ **B.** 6 rows
- ○ **C.** 8 rows

- D. 9 rows

Q16 (2 marks)

Here is part of a truth table for a 2-input AND gate:

A	B	Output
0	0	0
0	1	0
1	0	?
1	1	1

What value replaces the question mark?

- A. 0
- B. 1
- C. A
- D. B

Q17 (2 marks)

Here is part of a truth table for a 2-input OR gate:

P	Q	Output
0	0	?
0	1	1
1	0	1

1	1	1

What value replaces the question mark?

- A. 0
- B. 1
- C. P
- D. Q

Q18 (2 marks)

A heating system turns on when:
Temperature is LOW AND (Timer is ON OR Manual override is ON)
The temperature is LOW, the timer is OFF, and manual override is ON.
Will the heating turn on?

- A. No, because the timer is OFF
- B. No, because not all three conditions are met
- C. Yes, because temperature is LOW and manual override is ON
- D. Yes, because at least one condition is met

Q19 (2 marks)

Evaluate the Boolean expression: NOT (NOT A)
Where A = 1.

- A. 0
- B. 1

- **C.** -1
- **D.** 2

Q20 (2 marks)

A 2-input AND gate has a complete truth table.
How many rows have an output of 1?

- **A.** 1 row
- **B.** 2 rows
- **C.** 3 rows
- **D.** 4 rows

Q21 (2 marks)

A 2-input OR gate has a complete truth table.
How many rows have an output of 0?

- **A.** 1 row
- **B.** 2 rows
- **C.** 3 rows
- **D.** 4 rows

Q22 (2 marks)

Evaluate: (NOT A) OR B
Where A = 1 and B = 0.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** A
- ○ **D.** B

Q23 (2 marks)

A computer grants access when:
(Username is correct AND Password is correct) OR Admin override is enabled
A user enters a wrong password but admin override is enabled.
Will access be granted?

- ○ **A.** No, because the password is wrong
- ○ **B.** No, because username AND password must both be correct
- ○ **C.** Yes, because admin override is enabled
- ○ **D.** Yes, because at least two conditions are met

Q24 (3 marks)

Complete the truth table for the expression: A AND B AND C

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0

1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	?

What value replaces the question mark?

- **A.** 0
- **B.** 1
- **C.** 3
- **D.** ABC

Q25 (3 marks)

A logic circuit implements: (A OR B) AND (NOT C)
For which input combination is the output 1?

- **A.** A=0, B=0, C=0
- **B.** A=1, B=0, C=0
- **C.** A=0, B=0, C=1
- **D.** A=1, B=1, C=1

Q26 (3 marks)

A fire alarm system must sound when:
• Smoke is detected, OR
• Heat is detected AND the manual button is pressed
Which Boolean expression correctly represents this system?

- ○ **A.** Smoke AND Heat AND Button
- ○ **B.** Smoke OR Heat OR Button
- ○ **C.** Smoke OR (Heat AND Button)
- ○ **D.** (Smoke OR Heat) AND Button

Q27 (3 marks)

A logic circuit has the following truth table:

X	Y	Output
0	0	1
0	1	1
1	0	1
1	1	0

Which Boolean expression matches this truth table?

- ○ **A.** X AND Y
- ○ **B.** X OR Y
- ○ **C.** NOT (X AND Y)
- ○ **D.** NOT (X OR Y)

Q28 (3 marks)

Evaluate the Boolean expression: (A AND B) OR (B AND C) OR (A AND C)
Where A = 1, B = 0, C = 1.

- A. 0
- B. 1
- C. 2
- D. 3

Q29 (3 marks)

A security system uses three sensors: Motion (M), Door (D), and Window (W).
The alarm triggers when: M AND (D OR W)
Which scenario will NOT trigger the alarm?

- A. Motion detected, door opened, window closed
- B. Motion detected, door closed, window opened
- C. No motion, door opened, window opened
- D. Motion detected, door opened, window opened

Q30 (3 marks)

A logic circuit implements: A OR B OR C
In a complete 3-input truth table, how many rows have an output of 1?

- A. 1 row
- B. 3 rows
- C. 7 rows
- D. 8 rows

Total: **55** marks • Time: **30 min**

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